

# Mol-JEPA: what the compiler changes

Selected rewrites under a SMILES-only input contract

## 1 SPECIALISE THE INPUT

SMILES  $\rightarrow$  graph encoder

Extra modality inputs: absent



Omit their inactive encoders

Saves weights; no active work removed

**15.46M fewer parameters**

## 2 COMPOSE AFFINE MAPS

$$x \rightarrow Ax + a \rightarrow W(Ax + a) + b$$



$$x \rightarrow (WA)x + (Wa + b)$$

The consumer sees the narrow input.

Retain the original projection for residuals.

**5.50M fewer parameters**

## 3 STORE THE INTERACTION

$$(Qx_i)^\top (Kx_j)$$



$$x_i^\top (Q^\top K)x_j$$

Store the bilinear map when smaller.

Preserve biases, edge terms and each head.

**4.68M fewer parameters**

## RETAINED OUTPUTS

12 predicted embeddings · CLS · 13 latent embeddings · requested attentions

Schematic of selected projections, not a full architecture diagram. Float32 reassociation can change final bits.